The Anchor Accumulation Problem: Bounded Anchor Sets Recover Plasticity in Usage-Localized Elastic Weight Consolidation

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Abstract

Usage-localized elastic weight consolidation (DeLP) protects dendritic segments with per-neuron elastic anchors accumulated over tasks. Decomposing retention as $\bar{a}_T = \ell + \frac{T-1}{T}\,\mathrm{BWT}$ localizes the cost of long sequences in the *learning term* ℓ, not the architecture:/nanchors persist at full weight as an elastic drag on later learning. We call this the *anchor accumulation problem* and bound the stack at M entries. A FIFO budget recovers +4.5 and +9.0 points of final accuracy over unbounded DeLP at T=10 and T=20 (M=5; n=10 paired seeds; Wilcoxon $p = 0.002$); at M=2, 66.7% (+12.3 points). The budget is a distinct knob from the elastic weight λ: it controls the *age structure* of forgetting, λ its *amount*. Mass-keyed eviction criteria fail — they grandfather stale anchors and evict anchors at birth, losing −34.5 BWT — whereas age-first criteria are immune. Merging removes redundant inter-anchor cross terms: the penalty *value* drops one-sidedly by up to 64% (gradient unchanged) while accuracy stays indistinguishable from unbounded. The effect replicates on Fashion-MNIST and Split-CIFAR-100, where at T=20, M=2, five of six bounded policies beat unbounded (up to

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+5.0). A tuned online-EWC discount overtakes the budget at every grid cell: discounting is itself a soft bound, and soft and hard bounds obey one log-linear law in the protected-mass horizon ($R^2 = 0.99$) — so bounding anchor mass, not eviction, is the active ingredient. A no-protection baseline stays above every arm on Split-CIFAR-100; the residual is an auditable protection set.

Keywords: continual learning, elastic weight consolidation, catastrophic forgetting, dendritic computation, stability–plasticity dilemma

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1. Introduction

Elastic weight consolidation (EWC) [1] and its variants protect important parameters by penalizing departure from values that served previous tasks. The idea originates in the computational principles of synaptic memory consolidation [2], and a long line of work has since shaped it: synaptic intelligence [3] accumulates path integrals online, memory-aware synapses [4] estimate sensitivity without supervision, and context-dependent gating [5] makes the protection pathway itself task-conditional. Surveys of the field [6, 7] document both the breadth of these designs and how few of them are comparable under a common protocol — a gap our own audit of DeLP [8] ran into directly. DeLP [8] localizes this protection to dendritic compartments and weights each neuron's penalty by its measured usage under previous tasks. The audit of the method established that usage weighting survives a parameter-budget cap (+1.79, p=0.005 at twenty tasks) while the full dendritic chain adds little once baselines share the hidden activation.

The same audit exposed a structural cost: DeLP appends one anchor

set (posterior means and Fisher estimates) per task, so the elastic drag on optimization grows with the sequence length T. A decomposition of average retention,

$$\bar{a}_T \;=\; \ell \;+\; \frac{T-1}{T}\,\mathrm{BWT}, \tag{1}$$

separates the *learning term* ℓ (how well each task is acquired at its boundary) from the backward-transfer term. At ten tasks the losses of unbounded DeLP are concentrated in ℓ: anchors from early tasks persist as elastic drag long after the tasks they protect have been consolidated, and this is a property of the protection scheme, not of the architecture.

This paper treats the anchor stack itself as the object of study. We bound it at M entries and ask three questions: **(Q1)** Can the learning term be recovered by bounding, and at what stability cost? **(Q2)** Is the budget a redundant knob (a non-linear re-scaling of λ), or does it control something λ cannot? **(Q3)** If the stack must be bounded, *which* anchors should be evicted?

Contributions..

1. **Bounded anchor sets.** We introduce a family of eviction policies for anchor stacks and show that a FIFO budget recovers +4.5/+9.0 points of final accuracy at T=10/20 over unbounded DeLP (n=10 paired seeds, $t(9)$=21.4/19.1; Wilcoxon signed-rank p=0.002 in both cases), by recovering +5.4/ + 15.2 points of the learning term (Eq. 1). A budget of M=2 reaches 66.7% at T=10 (+12.3 over unbounded, n=10 paired seeds, Wilcoxon p=0.002), and the effect replicates on Fashion-MNIST (+13.6, n=10 paired seeds) and Split-CIFAR-100 (ResNet-18;

at T=20, M=2, five of the six bounded policies beat unbounded, up to +5.0 points, $t(9)$=7.1, Wilcoxon p=0.002; at M=5, merge, which trades a small BWT reduction for storage, is the exception).

2. **Two orthogonal knobs.** Through per-task accuracy profiles we show the budget controls the *age structure* of forgetting (which tasks degrade) while λ controls its total amount; the profiles are qualitatively different shapes, not re-scalings of one another (Fig. 2).
3. **A grandfathering failure mode.** Eviction criteria keyed to an anchor's current constraint mass systematically evict anchors *at birth* (mass ≈ 0 when the stack just changed) and preserve stale pre-budget anchors; BWT collapses to -34.5. Multiplicative age re-weighting does not fix this; an age-first, mass-second criterion does. Trace instrumentation of anchor stacks (Fig. 4) shows the mechanism directly.
4. **A compressed memory axis.** Fisher-cosine merging compresses the stack from T to M entries by removing exactly the cross terms between merged anchors: penalty instrumentation shows a one-sided drop (per-run maximum 18–64%) while the learned trajectory stays indistinguishable from unbounded, separating memory cost from plasticity.
5. **A mechanism-level boundary.** Sweeping online EWC's discount $\gamma \in \{0.5, 0.9, 1\}$ shows a tuned discount overtakes the budget on accuracy at matched forgetting, and that soft (discount) and hard (window) bounds collapse onto one log-linear law between the learning term and the protected-mass horizon: bounding accumulated anchor mass is the active ingredient.
6. **An auditable, policy-selectable protection set.** Because the bud-

get is a *set* of named anchors rather than a scalar discount, the protected mass is inspectable: which tasks are still protected, which were evicted, and by which policy, are all readable directly from the stack trace. A tuned online-EWC discount achieves the same accuracy level but hides this structure inside a single decay factor. The contribution is therefore a *measurable mechanism* (budget, not discount, is the active ingredient) and a *knob* (age structure vs. total mass), not a drop-in accuracy win over a tuned baseline.

2. Related work

Regularization-based continual learning.. EWC [1] places a quadratic penalty around posterior estimates of previous tasks; SI [3] accumulates path-integral importances online; MAS [4] estimates sensitivity unsupervised. Anchor accumulation is not specific to DeLP: any scheme that stores per-task quadratics pays storage $O(T)$ and, if all terms persist at full weight, an increasingly rigid objective. Online EWC [9] addresses the storage axis by collapsing the stack into a single running posterior; with a discount $\gamma<1$ its Fisher mass is geometrically bounded, which makes it a *soft* bound on the same accumulated mass we diagnose (Sec. 5.1). Bounded anchor sets are complementary: they keep per-task terms and bound their number.

Recent continual-learning methods in this journal.. Recent work in *Pattern Recognition* treats the stability–plasticity trade-off from several angles, and our work is complementary to all of it because it targets the *accumulation mechanism* of a regularizer rather than the network parameterisation. Forget to Learn [10] circumvents the conflict in continuous unsupervised domain

adaptation by active forgetting plus conflict-free distillation; TIPS [11] selects prompts at two granularities to rebalance stability and plasticity in prompt-based continual learning; a dual-learner with cumulative parameter averaging [12] pairs a plastic learner with a stable one; and LoRAC [13] constrains critical parameter changes under orthogonal LoRA tuning. These are all *architectural or replay-side* remedies. Ours is complementary: we ask what happens *inside* a regularizer that accumulates one anchor set per task, and we show that the accumulation itself — not the eviction rule, not the discount schedule — is what costs plasticity. We cite the rehearsal-side family [14] as the reference point that makes exemplar-free regularizer comparisons legible: a strong baseline of this kind forces one to state honestly in which regime a comparison holds.

Parameter isolation and budgeting.. PackNet [15] and HAT [16] budget capacity explicitly. Our contribution differs: the budgeted object is the *protection* state, not the network capacity, and eviction can be nearly free when the penalty is combined correctly (merge).

Dendritic computation.. The dendritic substrate is not decorative. Pyramidal neurons are multicompartmental systems in which local dendritic potentials shape integration at the soma [17, 18, 19], and these computations are measurable *in vivo* rather than merely modeled: dendritic activity in the cortex is multiplexed and globally modulated [20], and it computes abstract rules [21]. Reviews of dendritic computation [22, 23] and of its engineering exploitation [24, 25, 26, 27] make the relevant point for us: the compartment is a computational unit with its own timescales, so protecting it with

a task-wise snapshot is a substantive modeling choice rather than an implementation detail. Active-dendrite networks [28, 29] provide the architectural substrate; DeLP's audit [8] established which part of the dendritic chain carries the benefit.

Scope of this manuscript in relation to the author's precursor work.. Two earlier Zenodo preprints by the same author [8, 30] audited, respectively, which component of the dendritic chain carries the DeLP benefit and how that benefit distributes across input segments. Both are diagnostic studies of *where* the benefit originates; neither examines what happens to a stack of anchors as the task sequence grows. The present manuscript takes the opposite direction. Its subject is the cost that *any* stack-building scheme pays over a long sequence — the anchor accumulation problem — and its object is the repair for that cost: bounding the anchor stack at M entries, the eviction criteria that decide which anchors survive, and the resulting law relating soft and hard protection bounds. All experiments, analyses, figures, and text in the present manuscript are new; the precursor audits are cited as prior work and are not extended, reproduced, or reused here.

3. Background: DeLP and anchor accumulation

Setup.. A network is trained on a sequence of T tasks (permuted MNIST / Fashion-MNIST). DeLP freezes dendritic segments unused by the current task and adds, after each task t, an anchor set $\mathcal{A}_t = \{(\theta_t, F_t)\}$ (parameter snapshot and Fisher diagonal). The protection objective during task t is

$$\mathcal{L}_t(\theta) \;=\; \mathcal{L}_t^{\text{task}}(\theta) \;+\; \frac{\lambda}{2} \sum_{k<t} \sum_{i} F_i^{(k)} \, (\theta_i - \theta_i^{(k)})^2, \tag{2}$$

where the inner sum runs over the accumulated stack $\mathcal{A}_1, \ldots, \mathcal{A}_{t-1}$: the penalty is *unbounded* in T, both in storage and in constraint mass.

Where the cost lands.. Following [8], average retention decomposes as Eq. 1. At fixed λ=5000 the learning term of unbounded DeLP falls from 59.2 at T=10 to 37.7 at T=20 — a 21.5-point drop that compounds with sequence length, exactly the signature of drag that accumulates with every finished task; anchored identity Eq. 1 attributes the gap to ℓ, not BWT. We call this the *anchor accumulation problem*.

4. Bounded anchor sets

Budgeted stacks.. We replace the unbounded stack with a bounded one of size M. After appending the anchor for task t, if $|\mathcal{A}| > M$ an *eviction policy* removes or combines entries:

$$\mathcal{A} \leftarrow \mathsf{evict}_\pi(\mathcal{A} \cup \{(\theta_t, F_t)\},\, M)\,. \tag{3}$$

Policies.. Table 1 defines the six policies we compare. They span the design space along one axis: *what signal drives the choice* — age (fifo), chance (random), current constraint mass (redun), pairwise redundancy (merge), or combinations of age and mass (agew, agew2).

Instrumentation.. Each anchor carries an *origin* tag (the task that created it), and we record the surviving-origin set after every phase (`survivor_history`). This turns the eviction dynamics into a directly observable object (Fig. 4).

Table 1: Eviction policies for a stack over budget. Constraint mass $m_k = \sum_i F_i^{(k)}(\theta_i^{\text{now}} - \theta_i^{(k)})^2$ is measured at eviction time.

Policy	Key	Rule (when $\lvert\mathcal{A}\rvert > M$)
unbounded	—	never evict (original DeLP)
fifo	age	evict the oldest anchor
random	—	evict uniformly at random
redun	mass	evict $\arg\min_k m_k$
agew	mass × age	evict $\arg\min_k m_k \cdot (k{+}1)$, oldest $k{=}0$
agew2	age → mass	evict oldest anchor with below-median m_k; fallback fifo
merge	redundancy	merge the pair with most similar F (cosine): $F \leftarrow F_i + F_j$, $\theta \leftarrow$ weighted mean; repeat

5. Experiments

Protocol.. Permuted-MNIST and Fashion-MNIST, $T \in \{10, 20\}$; architecture, optimizer, permutation generation, and evaluation identical to the DeLP audit protocol [8]; λ=5000 unless stated. Every arm runs at the pipeline's *default* hidden-layer gain — the uncalibrated condition of the audit's calibration protocol [8] — and no arm is gain-tuned within this study; since the gain is shared, the confound the audit documented (a single gain scalar moves accuracy by more than the entire method) cannot differentially favor any arm, only shift all arms together. Paired comparisons use identical seeds. Budgets $M \in \{2, 5\}$ anchor the main grid; a seven-point dose–response sweep $M \in \{2, 3, 4, 5, 6, 8, 10\}$ (T=20, λ=5000; Sec. 5.2) fills the interior, and no M was tuned post hoc. Section 5.6 states the Split-CIFAR-100 protocol

Table 2: Paired fifo–unbounded deltas, M=5, λ=5000, n=10 seeds. $t(9)$ is the paired t statistic; W_p is the two-sided p-value of the Wilcoxon signed-rank test on the same paired deltas, and W_p^{SW} is the Shapiro–Wilk p-value on those deltas (normality of the paired difference, on which the t test depends).

Arm	Δacc	$t(9)$	W_p	W_p^{SW}	ΔBWT
fifo T=10	$+4.51 \pm 0.67$	21.4	0.002	0.024	-1.50 ± 0.49
fifo T=20	$+8.98 \pm 1.49$	19.1	0.002	0.992	-7.20 ± 1.97

separately (different backbone and protection pathway).

5.1. Bounding recovers plasticity (Q1)

Table 2: a FIFO budget at M=5 trades 1.5/7.2 points of BWT for +4.5/+ 9.0 accuracy. The mechanism is the learning term: ℓ rises by +5.4 and +15.2, exactly the recovery predicted if accumulated elastic drag were the binding constraint (Sec. 3); the grid in Table 3 shows the same signature for every bounded arm. Eq. 1 is an accounting identity, so the decomposition alone is not evidence; the causal claim rests on the eviction traces (Sec. 5.3) and on the λ-by-budget interaction (Sec. 5.2), which localize *when* the learning term moves — namely when anchors are actually retired. Corrected SI at T=10 (λ=100) reaches 40.3%.

Single-anchor references.. Because the diagnosis targets the *stack*, we also ran two protection schemes that cannot accumulate one. Both use the identical architecture, protocol, and λ grid (n=10 paired seeds): plain EWC, which keeps only the most recent task's Fisher and anchor (no accumulation, no running Fisher update — distinct from the original accumulated variant of [1]), and online EWC [9] with $\gamma \in \{0.5, 0.9, 1\}$, which collapses the

Table 3: Policy grid, M=5, λ=5000, mean ± sd over n=3 seeds. ℓ = learning term (mean accuracy at task boundaries). agew is bit-identical to redun, as predicted in Sec. 5.3.

T	Arm	BWT	acc	ℓ
10	unbounded	-5.5 ± 0.6	55.3	60.2
10	fifo	-6.6 ± 0.9	59.7	65.6
10	random	-9.1 ± 2.6	60.2	68.4
10	redun	-18.3 ± 0.8	53.6	70.1
10	agew	-18.3 ± 0.8	53.6	70.1
10	agew2	-8.4 ± 0.9	60.8	68.3
10	merge	-5.5 ± 0.6	55.3	60.3
20	unbounded	-4.4 ± 1.0	33.7	37.9
20	fifo	-10.2 ± 1.9	43.4	53.1
20	random	-13.5 ± 0.4	42.0	54.8
20	redun	-30.2 ± 0.4	36.3	65.0
20	agew	-30.2 ± 0.4	36.3	65.0
20	agew2	-16.5 ± 0.3	42.3	58.0
20	merge	-4.5 ± 0.9	33.6	37.9

stack into a single running posterior with geometrically discounted Fisher accumulation. All arms share the identical λ grid and are compared at each scheme's best grid value, so no scheme is disadvantaged by another's tuning. Plain EWC posts the highest final accuracy of any arm (68.3–71.1 at T=10, 56.4–59.4 at T=20, best at λ=1000) by maximizing ℓ (81–85): with no stack there is no accumulation — and no retention either, at 3–5× the forgetting of unbounded DeLP (BWT $-15.8/-25.7$ at its best λ vs. $-5.5/-4.4$). Under

the final-accuracy metric the learning-term dominance outweighs retention; a reader who weights BWT equally would rank it last. Online EWC at its best grid value (λ=1000) reaches 67.5/46.0: at T=10 it is statistically indistinguishable from DeLP-fifo at the same λ (67.4; paired $t(9) = -0.3$); at T=20 fifo is +3.5 ahead ($t(9)$=7.0). A collapsed posterior thus recovers most of the plasticity, but it posts neither the best long-sequence accuracy nor, by construction, any anchor accumulation problem to cure. Its forgetting is not unstructured: per-task forgetting (peak minus final accuracy; T=20, λ=1000, n=10) concentrates on older tasks for every arm, with drop–age correlations of −0.94 (plain), −0.79 (DeLP-fifo), −0.70 (online EWC), and −0.43 (unbounded). Sweeping the discount shows that the collapsed posterior does have an age knob after all: lowering γ to 0.9 and 0.5 raises ℓ from 51.1 to 56.6 and 76.1 (T=20, λ=1000), and at γ=0.5 online EWC overtakes DeLP-fifo at every grid cell we tested (paired +4.7 to +7.9, all $|t(9)| \geq 10.4$) at essentially matched BWT (−19.5 vs. −19.7). Discounting, however, is itself a bound: with geometrically decayed Fishers the accumulated mass converges to an effective horizon of $\approx 1/(1-\gamma)$ tasks, so γ<1 is a *soft* counterpart of the eviction window, and the two mechanisms turn out to be interchangeable points on one curve. Across all five bounded-mass arms at λ=1000, ℓ is log-linear in the protected horizon h (h=1 plain, h=2 for γ=0.5, h=5 for the window, h=10 for γ=0.9, h=20 unbounded): $\ell \approx 84 - 11 \ln h$ with R^2=0.99 — a fit that holds for every seed individually (per-seed $R^2 \geq 0.98$, slope -10.9 ± 0.6), replicates at T=10 (R^2=0.998 on the same four-arm fit excluding γ=0.9, whose ℓ sits 1.1 below the line), and whose ordering is monotone at every grid λ (mean-fit R^2 0.99/0.94/0.80 as λ tightens). Bounding ac-

cumulated anchor mass — not the mechanism that enforces the bound — is the active ingredient, and the soft bound is the stronger remedy on final accuracy in this regime. What the explicit window retains is auditability and policy-level choice: its protection set is discrete and named by the survivor trace; the forgetting profile it produces is the one the trace predicts (8.6 on recent tasks vs. 27.7 on the evicted oldest); and the eviction key can be swapped (age, mass) or the stack merged without touching the update rule. The object studied here remains the stack; the budget is one of two cures, and on this benchmark the simpler one wins.

5.2. Budget vs. λ: two orthogonal knobs (Q2)

Fig. 1 shows the Pareto plane; Fig. 2 the per-task accuracy profiles; Table 4 the λ interaction.

Dose–response.. A seven-point sweep at T=20, λ=5000 (n=10) shows the budget's effect is monotone across the whole interior, not an artifact of two extreme points: final accuracy $52.4, 48.9, 45.0, 42.4, 40.2, 36.5, 35.1$ for M=$2, 3, 4, 5, 6, 8, 10$, with BWT recovering $-24.5 \rightarrow -5.2$ as the window widens toward M=10; every M remains significantly above unbounded at the same λ=5000 (33.4; paired $t(9)$ from 36.0 at M=2 down to 9.1 at M=10, approaching the window's own limit).

The fifo advantage is robust to the choice of λ: even granting unbounded DeLP its best grid value at each T (λ=1000, the value at which unbounded is strongest at T=20), fifo retains a +10.9-point advantage there (and +3.5 at T=10). That the unbounded-optimal λ must drift downward as T grows is itself a symptom of accumulating rigidity: the protection budget has to

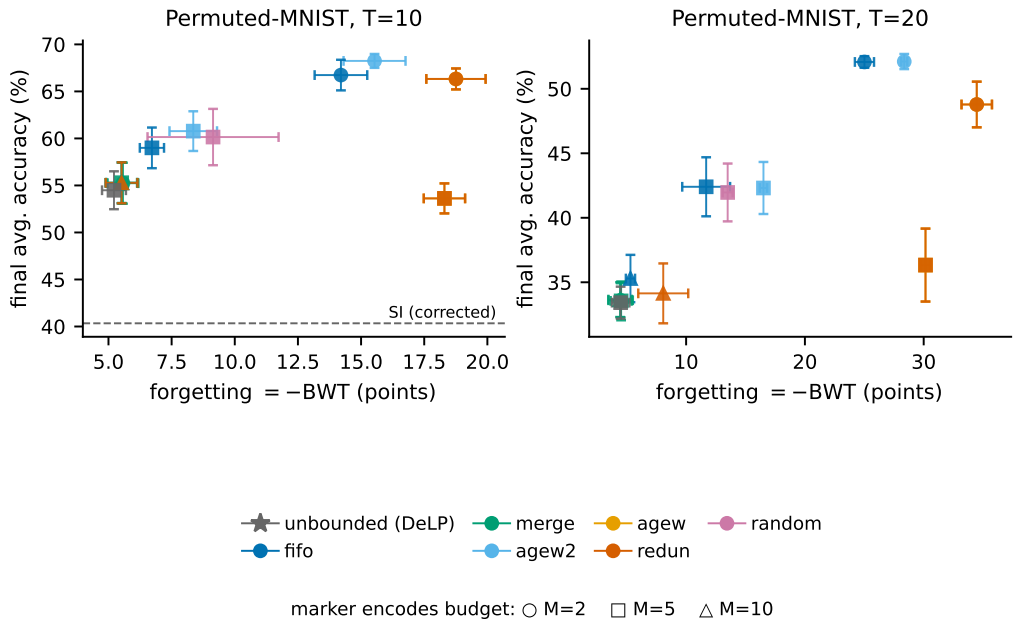


Figure 1: Accuracy–forgetting trade-off on Permuted-MNIST. Color encodes policy; marker encodes budget M. Error bars: sd over seeds (n=10 for unbounded/fifo M=5; else n=3). The smaller budget M=2 trades BWT for accuracy at both budgets in the same direction; merge coincides with unbounded at every M; redun sits dominated.

be relaxed sequence-wide to compensate for anchors that a bounded scheme would simply retire.

Both knobs move the system along a protection–plasticity spectrum, but they are not re-parameterizations of one another. At matched plasticity recovery, the two knobs produce *different age structures* of forgetting (Fig. 2): λ=1000 scales down protection uniformly — the profile keeps its steep age gradient (41.8-point spread, oldest tasks at 66–83%, newest at 41–50%) — while fifo M=2 redistributes forgetting onto the oldest tasks, flattening the profile (22.1-point spread). No value of λ produces the flat profile, and no

Table 4: λ interaction (M=5, n=10 paired seeds): the fifo advantage is positive at every λ and grows with T (all paired $t(9) \geq 11$, $p < 10^{-6}$).

λ	T	unbounded BWT/acc	fifo BWT/acc	Δacc
1000	10	−11.6/63.9	−10.8/67.4	+3.5
5000	10	−5.2/54.5	−6.7/59.0	+4.5
20000	10	−1.9/49.0	−4.0/53.1	+4.1
1000	20	−13.8/38.6	−19.7/49.5	+10.9
5000	20	−4.5/33.4	−11.7/42.4	+9.0
20000	20	−1.5/29.8	−6.6/37.3	+7.5

value of M produces the uniform scaling.

5.3. Which anchors to evict (Q3)

The grandfathering trap.. Mass-keyed eviction is not a straw man: "drop the anchor that constrains least" is the natural heuristic, homologous to low-sensitivity-first intuitions in pruning and in MAS-style importance estimates, and it is what a practitioner would try first. Its systematic failure is the point. Constraint mass m_k (Table 1) is measured at eviction time and is a function of staleness: anchors created before the budget first binds have $\theta^{(k)}$ far from θ^{now} and permanently large m_k; anchors created *after* the stack is full have $\theta^{(k)} \approx \theta^{\text{now}}$ and $m_k \approx 0$. Mass-keyed eviction therefore preserves exactly the stalest anchors and evicts new ones *at birth*: redun's stack freezes at $\{\mathcal{A}_1, \mathcal{A}_2\}$ (Fig. 4), the first two tasks keep strong protection (87/85% final accuracy at M=2), and tasks created later lose all protection within one phase (final accuracy 19–25%), driving BWT to −34.5 at T=20 despite the *highest* ℓ of any arm.

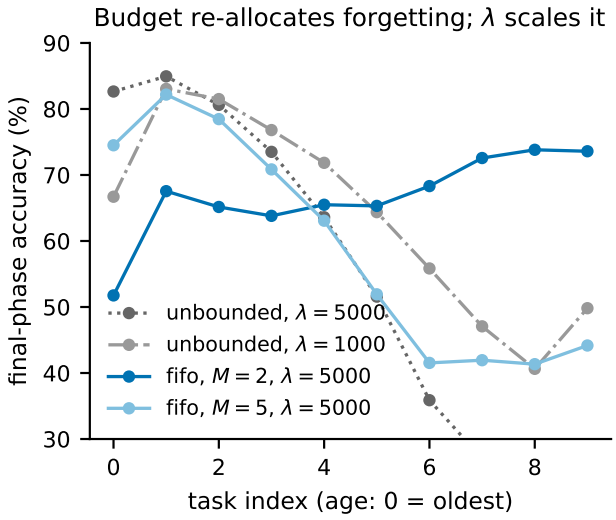


Figure 2: Final-phase accuracy per task (T=10, n=10). λ knobs (gray) translate the profile; budget knobs (blue) flatten it: forgetting is re-allocated to the oldest tasks.

Re-weighting mass by age does not help.. agew (mass $\times$ age-rank) is *bit-identical* to redun across all seeds and configurations: mass differences span orders of magnitude and swamp the age factor, so the argmin never flips. Any criterion with mass as the primary key inherits the trap.

Age-first criteria close the matrix.. agew2 (oldest anchor with below-median mass; fallback fifo) behaves like fifo ($\pm$1.5 points): at M=2 its stack is $\{\mathcal{A}_1, \mathcal{A}_T\}$ — the oldest anchor's mass is always above median and survives, everything in between dies at birth. Table 5 summarizes the resulting 2$\times$2: the primary key, not the secondary signal, determines whether a criterion works.

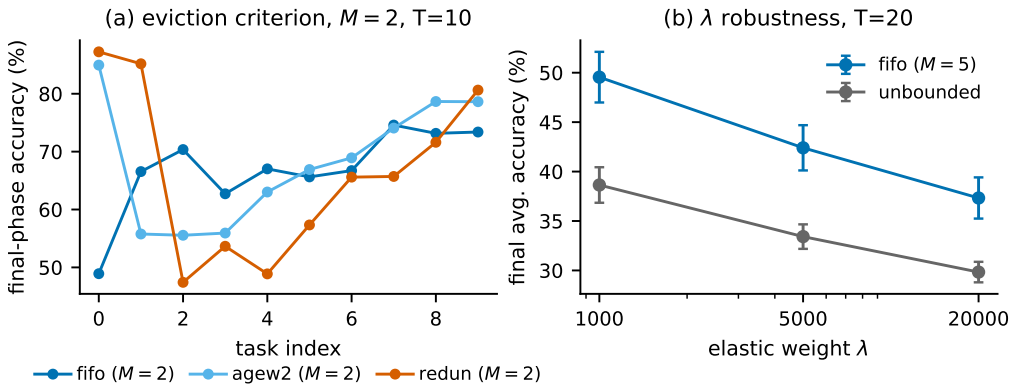


Figure 3: (a) Mass-keyed eviction (redun) grandfather-protects the first M tasks and erases the middle; age-first criteria (fifo, agew2) spread the cost. (b) The fifo advantage over unbounded holds at every λ.

5.4. Memory without behavior change: merge

Merging the most Fisher-similar pair removes exactly the cross term between the merged anchors. Penalty instrumentation over 40 runs ($T \in \{10, 20\}$, $M \in \{2, 5\}$, n=10; exactly $T-M$ merges per run) makes the change quantitative: the penalty's *value* drops one-sidedly — never increases — with per-run maximum relative drops of 43–45% at M=2 (worst case 64%) and 18–34% at M=5, and 77% of individual merges change that value. The change is *value-level only*: merging preserves the first-order quadratic regularizer (the gradient ∇_θ of the penalty is unchanged), shifting only the parameter-independent constant, which has zero effect on SGD. The removed mass is precisely the redundancy the merge detected; the accuracy trajectory nonetheless remains indistinguishable from unbounded at every M (Table 3) while storing $M \ll T$ anchors (-75% at T=20, M=5). Merge and fifo are complementary: merge buys storage at a measured, behaviorally invisible

Table 5: Criteria 2×2: primary key (rows) × secondary signal (columns). Only age-first criteria avoid the trap. T=10/T=20 BWT and accuracy, M=2 or M=5 as marked, n=3.

Primary key	Secondary	Outcome
mass	—	redun: grandfathering, BWT −30.2 (M5, T20)
mass	× age	agew ≡ redun (bit-identical)
age	—	fifo: sliding window, BWT −10.2 (M5, T20)
age	median mass	agew2 ≈ fifo (±1.5 pts)

penalty reduction but recovers no plasticity; fifo buys plasticity at a BWT price but no storage reduction.

5.5. Generality check: Fashion-MNIST

Fashion-MNIST, T=10, n=10 paired seeds: unbounded 35.1 ± 1.3; fifo M=5 38.1 ± 1.7 (+2.9, $t(9)$=11.5); fifo M=2 48.7 ± 2.5 (+13.6, $t(9)$=18.3). The budget effect replicates and amplifies as the budget shrinks.

5.6. Generalization to Split-CIFAR-100

Protocol.. Split-CIFAR-100, $T \in \{10, 20\}$: the 100 classes are split into T contiguous slices of $100/T$ classes each (10 classes/task at T=10, 5 classes/task at T=20; the code partitions a random permutation of the 100 classes into T blocks), ResNet-18 adapted to 32×32 inputs with GroupNorm in place of BatchNorm — running statistics are buffers that drift under later tasks and silently destroy protected tasks while the anchored weights themselves are held fixed, so they are excluded by design. Evaluation and training are task-incremental with a masked softmax head: each task computes its loss

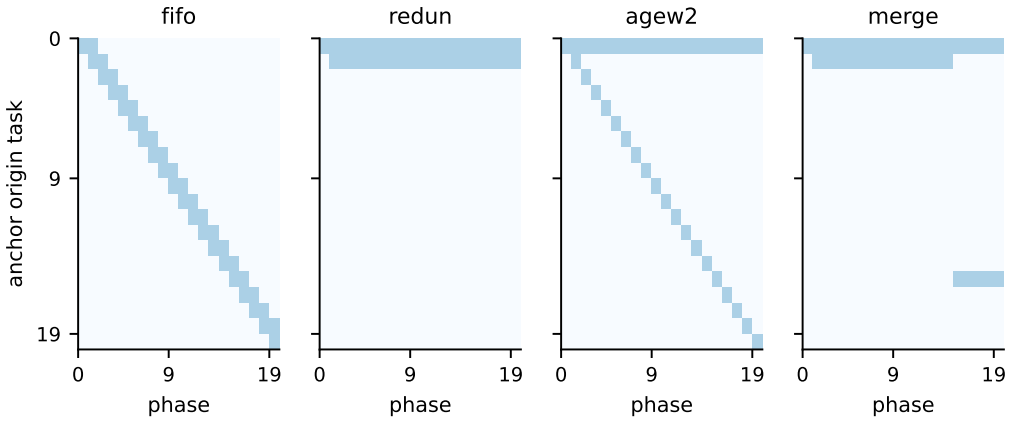


Figure 4: Anchor-stack composition over the task sequence (T=20, M=2, seed 0; identical for other seeds). Each column lists which anchors (by creation task) are alive at the end of each phase. fifo is a pure sliding window; redun freezes on the first two anchors (evicting every later anchor at birth); agew2 protects the oldest and the newest; merge partially grandfather-protects.

only over its own $100/T$ classes, so head rows of past tasks receive no gradient and are protected by construction; anchors therefore cover the backbone only, and forgetting can propagate exclusively through the shared trunk — exactly the pathway the anchor stack governs. Fisher diagonals are rescaled to unit global mean (one scalar over all parameters), making λ backbone-independent; gradients are clipped at norm 5. SGD, five epochs per task, λ=0.1 (chosen by an independent calibration sweep on unbounded DeLP only), n=10 seeds; eviction policies, origin instrumentation, and paired evaluation identical to the MNIST experiments. Unbounded DeLP never evicts, so its M is inert (final stack size T in every run); it is materialized once per T and serves as the reference for both budgets. All CIFAR comparisons are task-incremental (masked head); absolute accuracies are therefore not com-

Table 6: Split-CIFAR-100, λ=0.1: final accuracy, mean $\pm$ sd over n=10 seeds; parenthesized values are the paired difference Δacc against unbounded DeLP at the same seed (quoted in text with the paired $t(9)$ and the two-sided Wilcoxon signed-rank p-value).

	T=10		T=20	
Policy	M=2	M=5	M=2	M=5
unbounded	23.3 ± 2.6		29.8 ± 4.2	
fifo	24.8 ± 2.8 (+1.5)	23.5 ± 2.6 (+0.2)	$\mathbf{34.8 \pm 4.8}$ (+5.0)	32.2 ± 4.3 (+2.4)
random	24.7 ± 2.4 (+1.5)	22.4 ± 2.7 (−0.9)	33.4 ± 3.9 (+3.6)	32.2 ± 4.7 (+2.4)
redun	24.0 ± 2.5 (+0.7)	23.6 ± 2.0 (+0.3)	33.2 ± 4.7 (+3.4)	32.9 ± 3.3 (+3.1)
merge	22.3 ± 1.7 (−1.0)	21.8 ± 3.2 (−1.5)	30.8 ± 5.2 (+1.0)	29.3 ± 3.4 (−0.5)
agew	24.0 ± 3.8 (+0.7)	22.7 ± 3.4 (−0.6)	34.0 ± 4.3 (+4.2)	31.6 ± 3.7 (+1.8)
agew2	23.7 ± 2.6 (+0.4)	22.3 ± 3.3 (−1.0)	32.5 ± 3.7 (+2.7)	31.1 ± 3.4 (+1.3)

parable to the class-incremental Split-CIFAR-100 literature, and every claim in this section is a within-family, same-protocol paired comparison.

The budget effect strengthens with T on natural images.. At T=20 and the tightest budget (M=2), five of the six bounded policies outperform unbounded stacking (Table 6): fifo +5.0 ($t(9)$=7.1, Wilcoxon p=0.002), agew +4.2 (t=7.3, p=0.002), random +3.6 (t=9.2, p=0.002), redun +3.4 (t=4.0, p=0.006), agew2 +2.7 (t=2.7, p=0.037). At the looser budget (M=5) merge is the one policy that does not (−0.5), while the other five remain positive (+1.3 to +3.1); merge trades a small BWT reduction for storage and recovers no plasticity. At T=10 the effect is marginal, about +1.5 points at $t(9) \approx 2$, and no policy reaches significance under the Wilcoxon signed-rank test. The dose–response matches the mechanism: anchor drag accumulates with sequence length, so the benefit of bounding it should grow with T. As

in Sec. 5.1, the gain lands in the learning term: fifo M=2 raises ℓ from 31.4 to 36.8 while BWT moves only from -1.7 to -2.1 — the budget buys plasticity, the trade Eq. 1 predicts. The decomposition itself is exact on these runs: over all 260 CIFAR records, $|\bar{a}_T - (\ell + \frac{T-1}{T}\mathrm{BWT})|$ never exceeds 5×10^{-3}, i.e. the rounding of two-decimal reporting.

A no-protection reference.. λ=0 (plain SGD, same protocol, n=10 paired seeds) reorders the picture and must be reported: 27.0 ± 1.7 at T=10 and 36.9 ± 3.4 at T=20 — above every DeLP arm, including fifo M=2 (+2.1, $t(9)$=1.5, not significant). The decomposition explains why: protection roughly halves the learning term (ℓ is 25.7 and 31.4 at T=10/20 for unbounded DeLP, vs. 46.1 and 64.3 at λ=0), while in this regime retention is nearly free to forgo — even the large BWT of the unprotected arm ($-21.3/-28.9$) costs less than the ℓ it gains. Under task-incremental evaluation with a fixed masked head, trunk features are shared across tasks and trunk drift is cheap; regularization-based protection at any λ is then a net tax on accuracy. The budget effect remains real within the protected family, but on this benchmark bounding is necessary, not sufficient: protection itself must be checked against a no-protection reference before the mechanism is credited.

Stack traces replicate bit-for-bit.. At T=20, M=2, all ten seeds agree: fifo's stack is the sliding window $\{\mathcal{A}_{T-1}, \mathcal{A}_T\}$; redun freezes on $\{\mathcal{A}_1, \mathcal{A}_2\}$ (grandfathering); agew inherits the same freeze (eviction-time mass swamps the age factor); agew2 holds $\{\mathcal{A}_1, \mathcal{A}_T\}$ in 6/10 seeds (the age-primary hybrid); merge grandfather-protects partially. The mechanism isolated on Permuted-MNIST (Fig. 4) is not an artifact of that domain (Fig. 5).

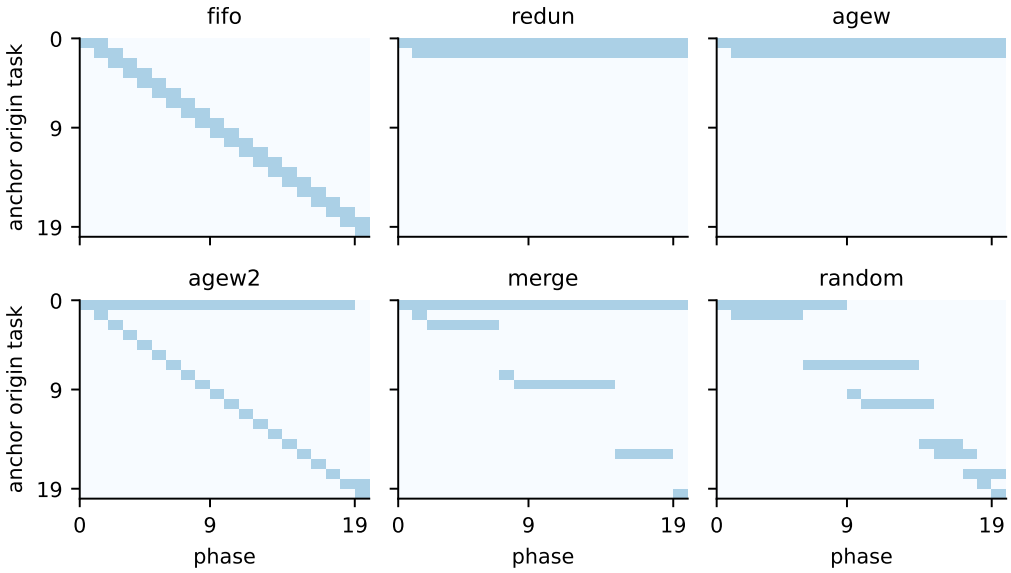


Figure 5: Anchor-stack composition on Split-CIFAR-100 (T=20, M=2, seed 0; all seeds agree on the final sets). Same rendering as Fig. 4. fifo is a sliding window; redun and agew freeze on the first anchors (grandfathering); agew2 pairs the oldest with the newest; merge partially grandfather-protects; random walks down.

A boundary condition on the mass-keyed failure.. Unlike on MNIST, the *secondary* effect — *which* criterion evicts — is weak here: fifo − redun is +1.6 ($t(9)$=2.0) at T=20, M=2, and agew2 − redun is slightly negative. The reason is structural: under masked-CE training, forgetting propagates only through the shared trunk, whose low layers are described similarly by every task's Fisher diagonal; even the stale anchors $\{\mathcal{A}_1, \mathcal{A}_2\}$ constrain features later tasks still use, so grandfathering them costs little. Mass-keyed eviction fails when anchors describe task-specific structure that goes stale; the primary effect of bounding (recovering plasticity) survives, while the relative cost of grandfathering scales with how much low-level structure tasks share

— a testable prediction for domain-incremental sequences, where the gap should re-open.

6. Discussion

Why bounding works.. Eq. 1 says total retention is learning plus transfer. Unbounded stacks tax learning with stale constraints. A FIFO window keeps only the freshest M constraints, whose $\theta^{(k)}$ are near θ^{now} and whose Fisher mass is relevant to the current basin: the penalty constrains what actually needs constraining.

The mass signal is not useless — it is mis-keyed.. redun asks "which anchor constrains least *now*?"; the honest answer at eviction time is "the one just created", which is exactly the wrong anchor to drop. agew2 shows mass adds marginal, unstable value as a secondary signal. The information structure of the problem makes age the only reliable primary key: eviction-time mass confounds relevance with staleness.

Limitations.. (1) Task-incremental sequences (permuted MNIST, Fashion-MNIST, Split-CIFAR-100 with a masked head); domain-incremental settings, where tasks share nearly all low-level structure, may shrink the age-over-mass gap further (Sec. 5.6). (2) $T \leq 20$; the ℓ-recovery trend grows with T but the BWT cost does too. (3) All MNIST arms share an uncalibrated hidden gain, as in the audit; the CIFAR study uses its own protocol with λ calibrated on unbounded DeLP only; conclusions are within-family comparisons. (4) Merge's penalty change is measured by instrumentation (Sec. 5.4), not bounded formally; a closed-form error bound is future work. (5) On

Split-CIFAR-100 the no-protection reference outperforms every protection arm (Sec. 5.6): task-incremental evaluation makes forgetting cheap, so the anchor pathway is a net tax there and the budget result is internal to protected arms. (6) The three benchmarks are the standard continual-learning benchmarks of the regularizer literature, not large-scale vision models: we do not evaluate ImageNet-scale class-incremental streams, so the magnitude of the anchor-accumulation cost in that regime is unmeasured. We regard the diagnosis, not the repair, as the transferable claim here — the accumulation law of Sec. 5.1 follows from the regularizer's form rather than from the task distribution, and should therefore scale; establishing the corresponding magnitudes at scale is the next step, and the instrumentation used here (anchor-stack survivor histories, per-run penalty traces) is designed to be portable to that setting.

7. Conclusion

Bounding the anchor stack converts the diagnosed anchor accumulation problem into a control knob: FIFO eviction recovers the learning term at a declared BWT cost, the budget's effect is structurally different from λ's, age is the only sound eviction signal, and merge decouples memory from behavior. The effects replicate across three benchmarks — permuted and class-incremental, MLP- and ResNet-based — with the age-over-mass advantage bounded by a stated boundary condition on task relatedness, and the protection pathway itself bounded by a no-protection reference on Split-CIFAR-100. Single-anchor schemes cannot accumulate the problem by construction, and a discounted posterior — itself a soft bound, its mass converging to a horizon

of $\approx 1/(1-\gamma)$ tasks — recovers plasticity through the same law: across soft and hard bounds the learning term falls log-linearly with the protected-mass horizon at the calibrated λ (monotone at every grid value), and a tuned discount overtakes the window on accuracy at matched forgetting. Bounding accumulated anchor mass, not the eviction mechanism, is the active ingredient; the explicit window is the variant whose protection set is auditable and whose policy is a choice. Protection schemes for continual learning should ship with bounded — or at least mergeable — anchor state whenever protection is warranted at all; the Split-CIFAR-100 reference (Sec. 5.6) shows that the warrant itself is regime-dependent and must be checked first.

Appendix A. Hyperparameters and reproducibility

Statistics.. All comparisons between two arms are *paired by seed*: each seed supplies one run per arm under an identical protocol, and every reported Δ is the per-seed difference. We report the paired t statistic $t(9)$ together with the two-sided p-value of the *Wilcoxon signed-rank* test on the same paired differences, and we report the Shapiro–Wilk p-value on those differences so that the normality assumption underlying the t test is visible rather than implicit. The choice to report both is deliberate. Whether a benchmark's headline number supports a claim depends on the variability of the metric under resampling, not only on its mean. Several authors have argued this point for generative evaluation [31, 32], for metric learning [33], for deep reinforcement learning [34], and, closest to our setting, for reproducibility in machine-learning scholarship as a whole [35]. A single significance test applied to ten seeds is exactly the setting in which those critiques bite: with

ten paired samples the parametric t test has assumptions that the data may not satisfy, while the distribution-free signed-rank test is valid at this sample size by construction. We therefore report both and let the reader see where they diverge. The two tests agree on every headline comparison: the FIFO budget beats unbounded at M=2 and M=5, T=10 and T=20, with Wilcoxon p=0.002 (the smallest attainable two-sided value at n=10 paired samples) in all four cases. One case deserves the explicit note: at M=5, T=10 the paired differences depart from normality (Shapiro–Wilk p=0.024), so the t test there rests on an assumption the data do not satisfy, whereas the distribution-free Wilcoxon result (p=0.002) does not. We therefore treat the Wilcoxon value as the primary significance statement throughout, and the t statistic as a secondary, parametric cross-check. The significance threshold is α=0.05 throughout, with no multiplicity correction: the grid in Table 6 contains twelve policy×budget cells and is read as a descriptive map of the policy landscape, with per-cell p-values reported so that the reader can judge each claim individually. All p-values above are recomputed from the raw per-seed records by `audit/audit_nonparametric.py`, which is included in the accompanying package.

Permuted-MNIST / Fashion-MNIST (Secs. 5 to 5.5).. Architecture as in the audit [8]: a 784–256 fully connected trunk whose pre-activation is modulated by K=T dendritic segments (linear context-to-segment map, winning-segment readout), hidden activity sparsified by k-WTA with k=64 = $N/4$, linear 10-way head. Adam, lr 10^{-3}, batch 256, two epochs and 10,000 samples per task; permutation matrices drawn once per seed and shared by all arms. Diagonal Fisher and usage statistics from 300 and 500 samples per task

respectively (max-log-probability Fisher variant); λ_{DeLP}=5000, corrected-SI reference λ=100. Paired arms use seeds 0–9; the policy grid seeds 0–2 (Table 3); λ grid as in Table 4. Every run seeds `torch.manual_seed` before model construction. Code and per-seed JSON records (including `survivor_history` and `acc_matrix`) are released with the repository.

Split-CIFAR-100 (Sec. 5.6).. ResNet-18 adapted for 32×32 inputs (3×3 stem, stride 1, no max-pool) with all BatchNorm layers replaced by GroupNorm (32 groups). SGD, momentum 0.9, weight decay 5×10^{-4}, lr 0.05, batch 128, five epochs per task. Masked cross-entropy over the task's $100/T$ classes; task-incremental evaluation over the same classes. EWC penalty over backbone parameters only, λ=0.1; diagonal Fisher estimated from 50 single-sample draws of the squared gradient of the max log-softmax on the finished task, rescaled to unit global mean; gradient clipping at max norm 5. Tasks are contiguous $100/T$-class blocks of a fixed random permutation of the 100 classes (generator seed 100+seed, matching the released code); seeds 0–9 control init, shuffle, and eviction randomness. Per-job crash-safe checkpoints and the full JSON records (including `survivor_history` and `acc_matrix`) are released with the code.

Appendix B. Full per-task accuracy matrices

Final-phase accuracy (%) per task, Permuted-MNIST, λ=5000, M=5: mean over seeds (n=10 for unbounded and fifo, n=3 for random/redun/merge). These underlie Fig. 3a and the grandfathering analysis of Sec. 5.3.

Table B.7: T=10.

task	unbounded	fifo	random	redun	merge
1	82.3	74.3	83.0	83.5	82.3
2	84.6	82.1	82.9	85.5	84.6
3	81.7	80.1	79.3	82.1	81.7
4	72.6	70.5	71.9	73.3	72.6
5	65.7	66.2	62.5	66.3	65.7
6	52.7	53.8	38.0	18.7	52.7
7	36.1	41.9	49.2	22.4	36.0
8	29.3	44.8	39.9	20.8	29.3
9	24.6	42.9	42.3	25.4	24.9
10	23.2	40.3	52.3	58.2	23.1

CRediT authorship contribution statement

Ranxi Zhou: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing — original draft, Writing — review & editing, Visualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All code, experiment scripts, and the complete JSON records of every run (including per-phase accuracy matrices and anchor-stack survivor histories) are publicly available at `https://github.com/zhouranxi/delp-anchor-budget`. This manuscript and its full reproducibility package — including the audit scripts that re-derive every quoted number from those records — are archived on Zenodo (doi:10.5281/zenodo.23668522). The precursor audit study is archived at doi:10.5281/zenodo.23572226 and the companion segment-specialization audit at doi:10.5281/zenodo.23637972; the underlying dataset and code release is doi:10.5281/zenodo.23639561.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used AI-assisted language-editing tools in order to improve the grammar, clarity, and consistency of the manuscript text. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article. The use of AI tools did not extend to any scientific content: all experimental design, analysis, figures, and results were produced and verified by the author.

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Table B.8: T=20.

task	unbounded	fifo	random	redun	merge
1	80.6	67.4	70.7	84.4	80.6
2	83.8	73.3	78.6	83.4	83.8
3	79.6	68.9	69.6	78.3	79.7
4	70.8	59.2	56.4	69.3	70.5
5	60.6	52.8	48.4	61.1	60.6
6	52.5	41.8	46.1	11.6	51.8
7	36.7	34.6	38.1	16.3	36.6
8	30.5	38.1	31.7	16.9	30.2
9	26.2	37.9	36.7	17.4	26.2
10	17.5	29.9	35.4	13.5	17.4
11	14.5	31.8	21.4	14.8	14.4
12	10.9	40.3	30.7	14.9	11.0
13	14.8	32.4	28.1	19.4	14.7
14	11.8	34.4	28.8	20.2	11.6
15	16.2	36.7	30.0	25.1	16.2
16	18.3	39.5	42.3	30.9	18.5
17	14.1	41.3	38.0	23.7	13.9
18	9.8	35.4	31.2	30.4	9.7
19	9.6	30.4	36.0	35.0	9.3
20	14.6	41.7	41.2	60.1	14.4